

## Experiment 2

# Designing and Implementing ETL Processes using Microsoft Power BI

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**Name:** Soumik Roy**UID:** 24BAI71017**SUB:** Business Intelligence

### Aim

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To design and implement an end-to-end ETL (Extract, Transform, Load) pipeline using Microsoft Power BI's Power Query Editor, enabling the extraction of data from a real-world business dataset, application of transformation logic, and loading of cleansed data into a unified analytical model for reporting and decision-making.

### Tools / Software Required

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1. Microsoft Power BI Desktop
2. Global Superstore Dataset.xlsx
3. Power Query Editor (Built into Power BI Desktop)
4. DAX (Data Analysis Expressions)

### Theory

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ETL (Extract, Transform, Load) is one of the most important processes in Business Intelligence. It enables organizations to collect data from source systems, transform it into a clean and consistent format, and load it into a centralized environment for analysis.

#### Extract Phase:

The extraction phase involves collecting data from source systems. In this experiment, data is extracted from the Orders, Returns, and People worksheets available in the Global Superstore dataset.

#### Transform Phase:

The transformation phase includes data cleaning, handling missing values, correcting data types, creating derived columns, filtering invalid records, and integrating multiple datasets.

#### Load Phase:

The transformed data is loaded into the Power BI data model where relationships can be established and analytical measures can be created using DAX.

Power Query Editor serves as the ETL engine in Power BI and records each transformation step as an Applied Step, making the workflow transparent and reproducible.

## Conduct

### Part A: Extract – Connect to Dataset

**Step 1:** Download the Global Superstore Dataset from Kaggle.

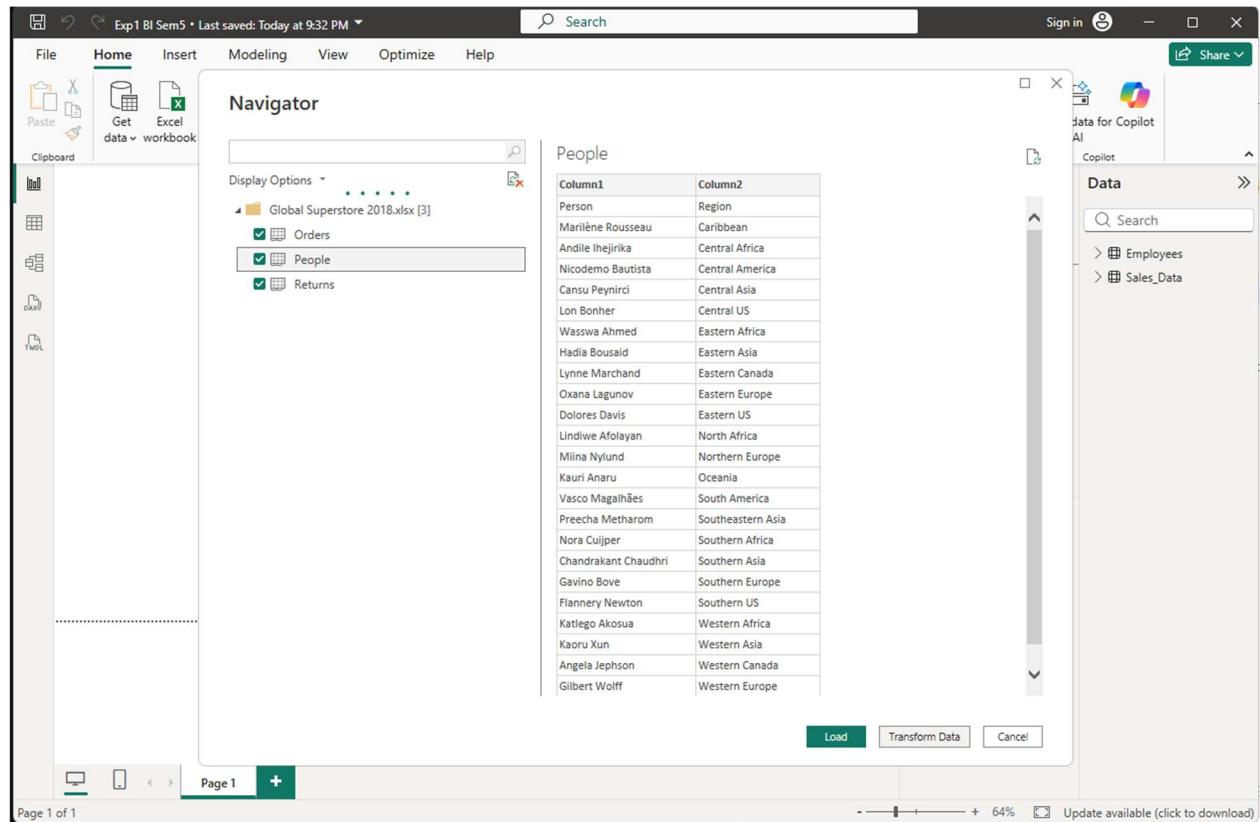
| 1 selected                   |              |               |           |      | 80% of storage used. If you run out, you can't create, edit, and upload files. Get 30 GB for ₹15 for 3 months ₹59. |  |  |  |  | Clean up space | Get offer | × |
|------------------------------|--------------|---------------|-----------|------|--------------------------------------------------------------------------------------------------------------------|--|--|--|--|----------------|-----------|---|
| Name                         | Owner        | Date modified | File size | Sort |                                                                                                                    |  |  |  |  |                |           |   |
| BI Worksheet Submission Form | ashiq.e20283 | Aug 10        | 32 KB     |      |                                                                                                                    |  |  |  |  |                |           |   |
| Employee_DB.xlsx             | ashiq.e20283 | Jul 17        | 8 KB      |      |                                                                                                                    |  |  |  |  |                |           |   |
| EXP 1.docx                   | ashiq.e20283 | Jul 16        | 14 KB     |      |                                                                                                                    |  |  |  |  |                |           |   |
| EXP 2.docx                   | ashiq.e20283 | Aug 10        | 10 KB     |      |                                                                                                                    |  |  |  |  |                |           |   |
| Global Superstore 2018.xlsx  | ashiq.e20283 | Jul 23        | 6.9 MB    |      |                                                                                                                    |  |  |  |  |                |           |   |
| Sales_Data.csv               | ashiq.e20283 | Jul 17        | 8 KB      |      |                                                                                                                    |  |  |  |  |                |           |   |
| Transactions.sql             | ashiq.e20283 | Jul 17        | 11 KB     |      |                                                                                                                    |  |  |  |  |                |           |   |

**Step 2:** Open Power BI Desktop.

Home > Get Data > Excel Workbook.

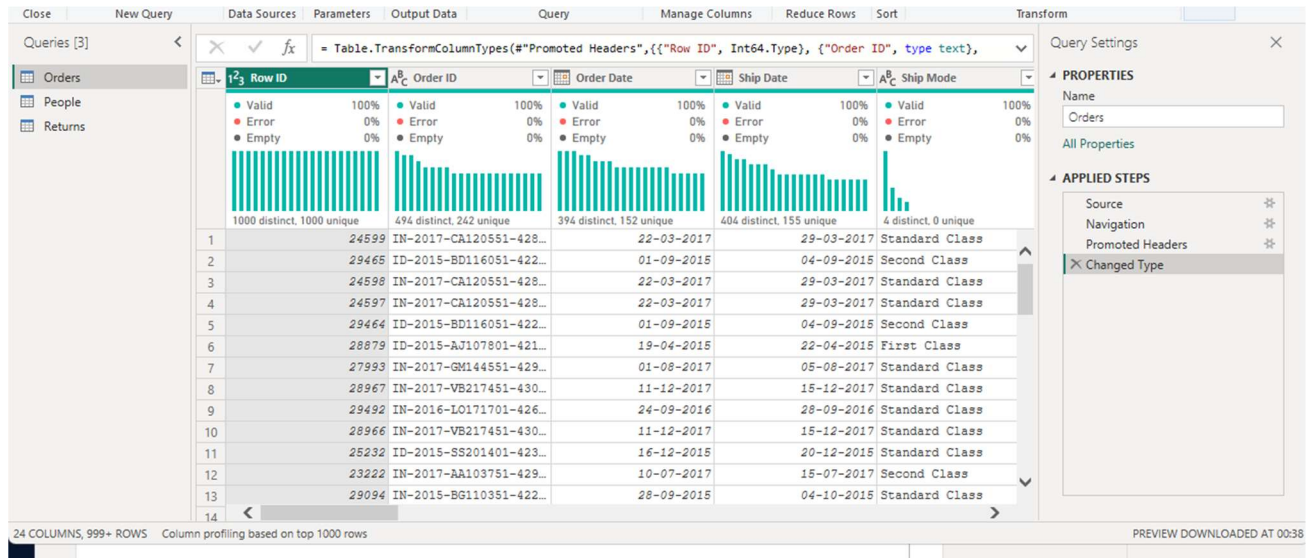
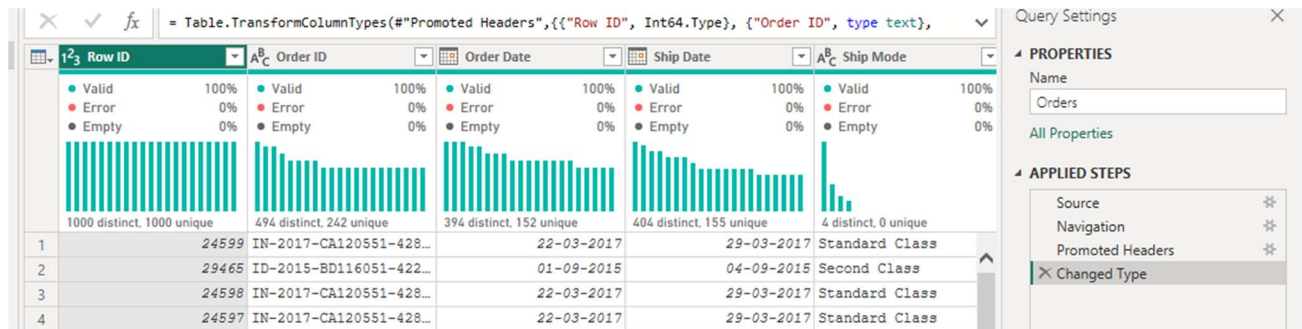
Select Global Superstore Dataset.xlsx.

Click Transform Data.



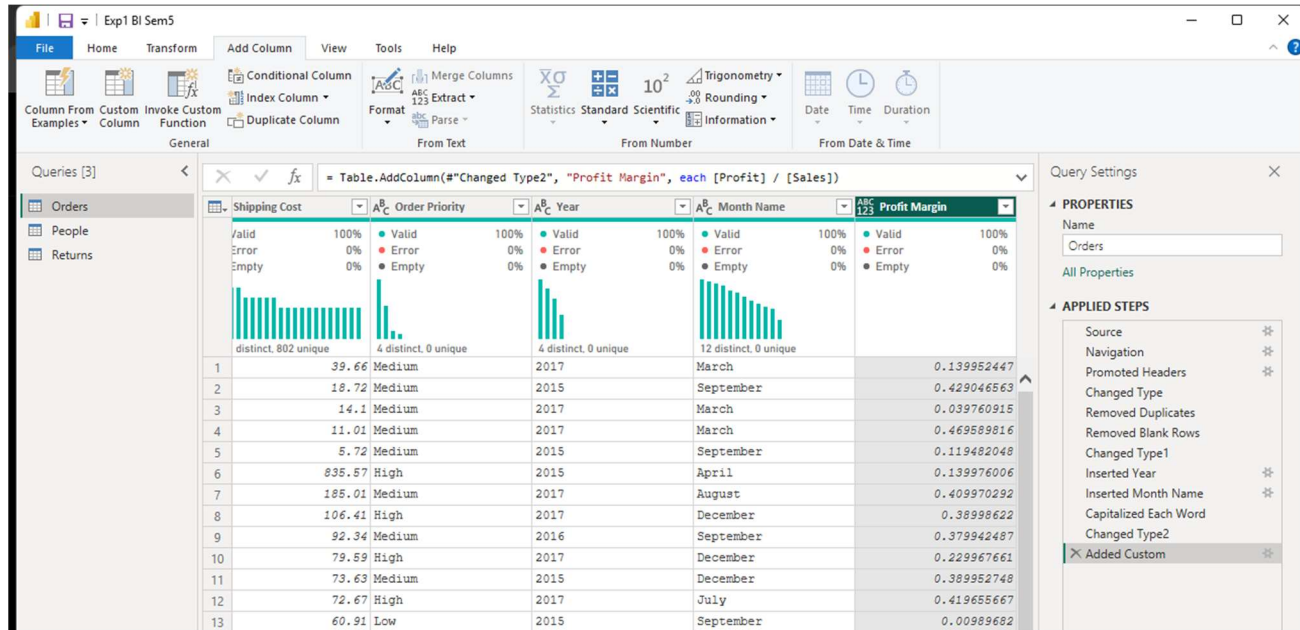
**Step 3: Import the following worksheets:**

- Orders
- Returns
- People

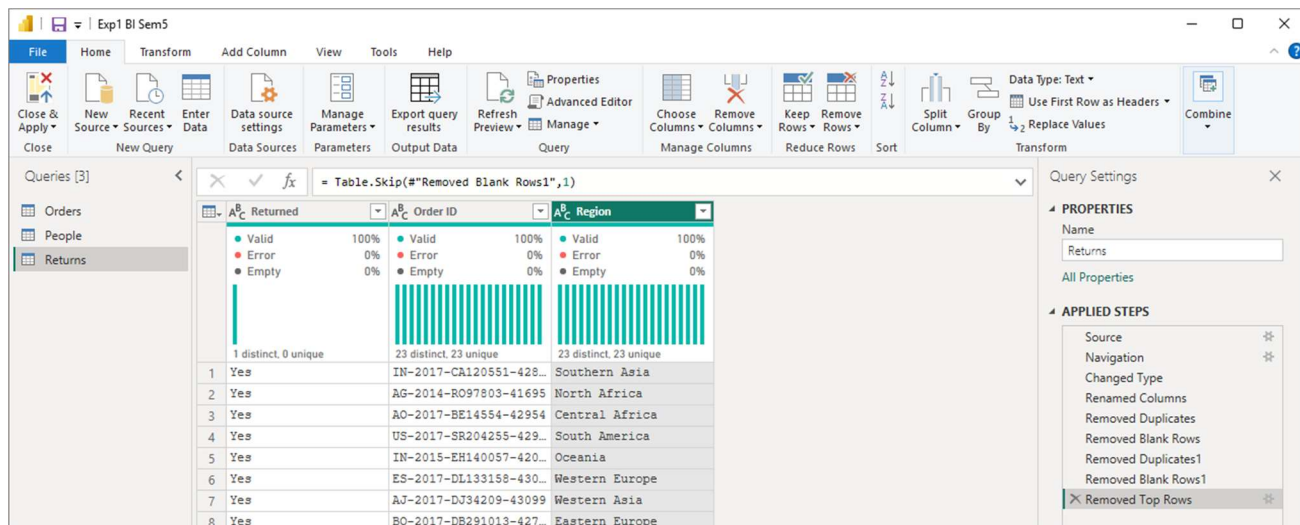
**Part B: Data Profiling in Power Query Editor****Step 4: Open Power Query Editor.**

**Step 5:** Apply the following transformations to the Orders table:

- Remove duplicate records
- Remove blank rows
- Verify and correct data types
- Rename columns where required
- Extract Year and Month from Order Date
- Create a custom Profit Margin column
- Standardize Region and Segment values

**Step 6:** Transform the Returns table:

- Remove duplicate Order IDs
- Check for missing values
- Standardize Returned field values



**Step 7: Transform the People table:**

- Validate regional assignments
- Remove unnecessary columns
- Verify consistency of region names

Query: Table.SelectRows(#"Removed Duplicates", each not List.IsEmpty(List.RemoveMatchingItems...

| Person | Region             |
|--------|--------------------|
| 1      | Marilène Rousseau  |
| 2      | Andile Ihejirika   |
| 3      | Nicodemus Bautista |
| 4      | Cansu Peynirci     |
| 5      | Lon Bonher         |
| 6      | Wasswa Ahmed       |

Query Settings: Name: People, All Properties, APPLIED STEPS: Source, Navigation, Changed Type, Renamed Columns, Removed Top Rows, Removed Duplicates, **Removed Blank Rows**.

**Step 8: Merge Orders and Returns tables using Order ID.**

Query: Table.NestedJoin(#"Added Custom", {"Order ID"}, Returns, {"Order ID"}, "Returns", ...)

| Quantity | % Discount | 1.2 Profit | 1.2 Shipping Cost | Order Priority |
|----------|------------|------------|-------------------|----------------|
| 1        | 0.00%      | 0.00%      | 102.42            | Medium         |
| 2        | 0.00%      | 0.00%      | 104.49            | Medium         |
| 3        | 0.00%      | 0.00%      | 13.77             | Medium         |
| 4        | 0.00%      | 0.00%      | 79.68             | Medium         |
| 5        | 0.00%      | 0.00%      | 24.36             | Medium         |
| 6        | 0.00%      | 0.00%      | 647.55            | High           |
| 7        | 0.00%      | 0.00%      | 848.7             | Medium         |
| 8        | 0.00%      | 0.00%      | 356.58            | High           |
| 9        | 0.00%      | 0.00%      | 443.94            | Medium         |
| 10       | 0.00%      | 0.00%      | 118.2             | High           |
| 11       | 0.00%      | 0.00%      | 564.48            | Medium         |

Query Settings: Name: Orders, All Properties, APPLIED STEPS: Source, Navigation, Promoted Headers, Changed Type, Removed Duplicates, Removed Blank Rows, Changed Type1, Inserted Year, Inserted Month Name, Capitalized Each Word, Changed Type2, Added Custom, **Merged Queries**.

**Step 9: Review all Applied Steps and click Close & Apply.**

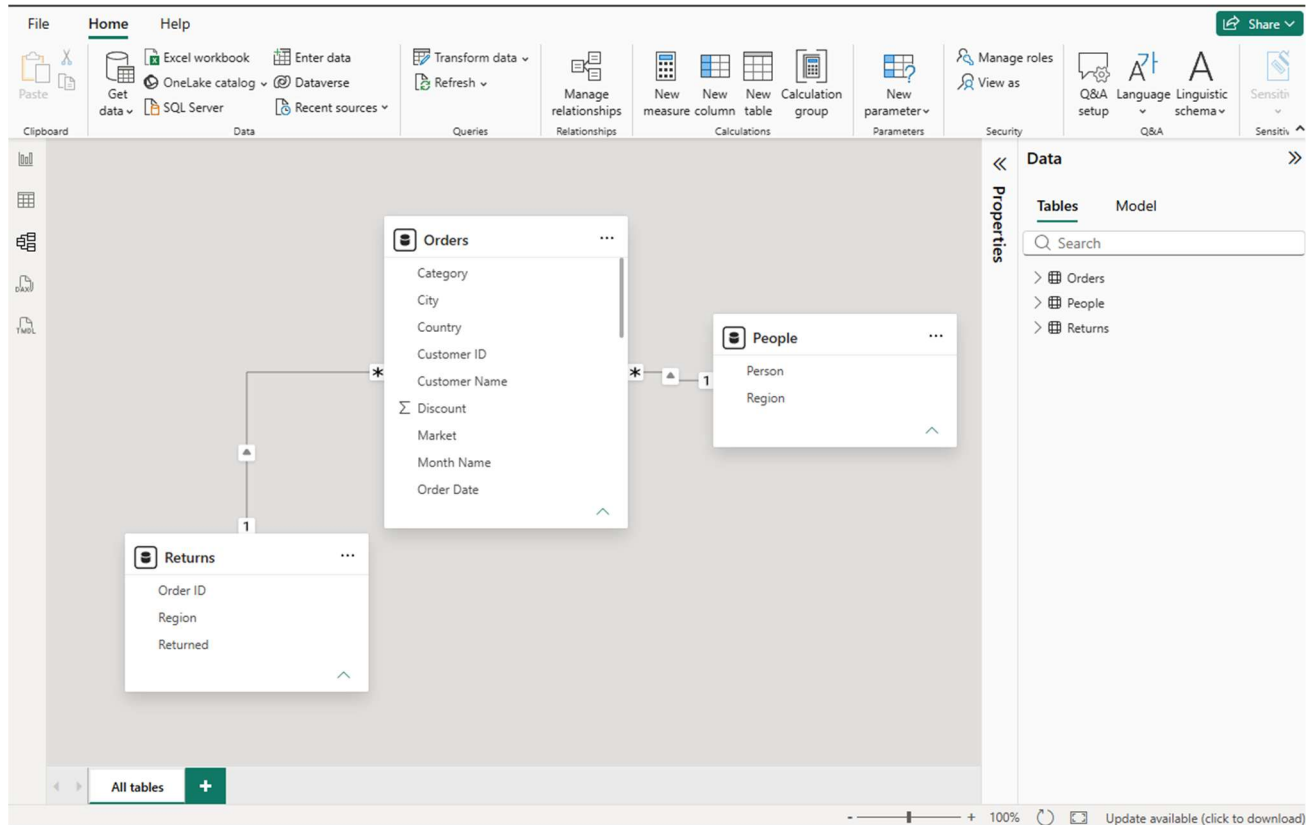
## Part C: Load – Build Data Model

### Step 10: Navigate to Model View

### Step 11: Create relationships:

Orders ↔ Returns (Order ID)

Orders ↔ People (Region)





**Step 12:**

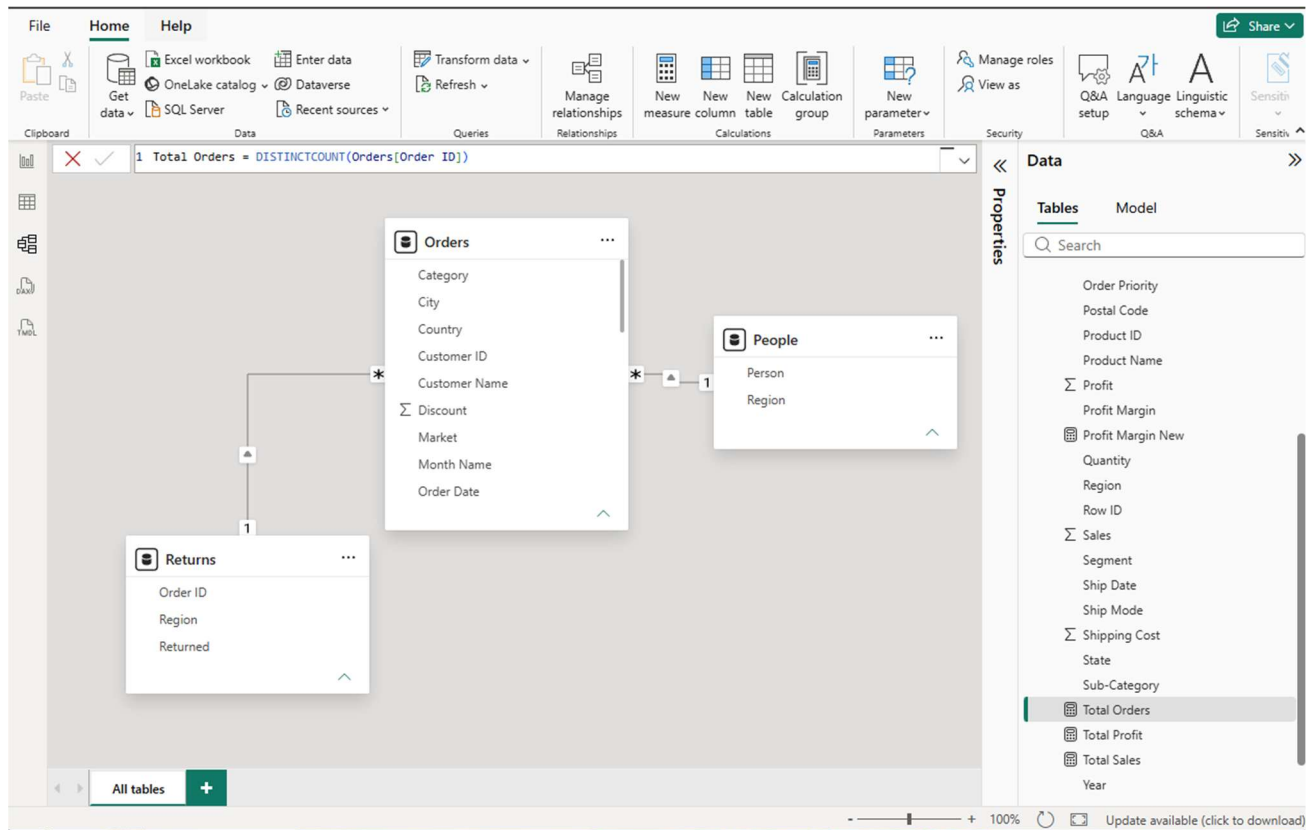
Create DAX measures:

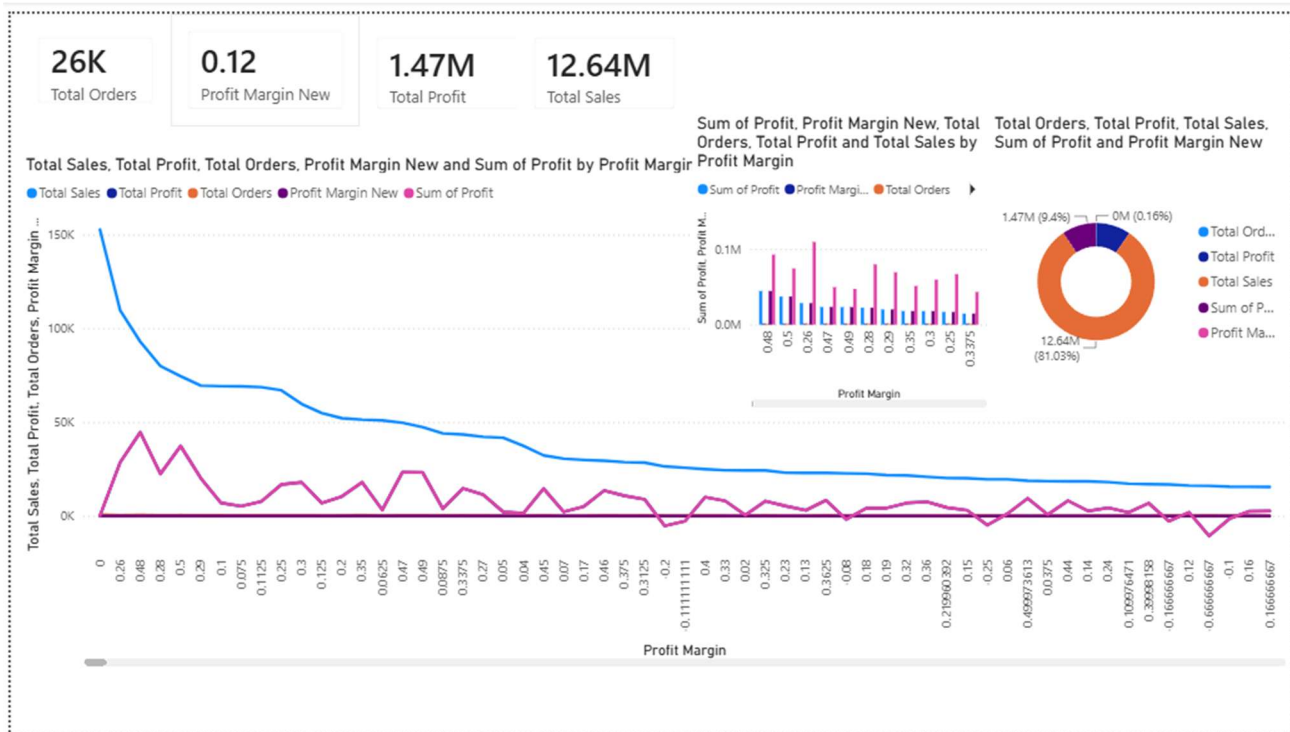
Total Sales = SUM(Orders[Sales])

Total Profit = SUM(Orders[Profit])

Profit Margin = DIVIDE(SUM(Orders[Profit]),SUM(Orders[Sales]))

Total Orders = DISTINCTCOUNT(Orders[Order ID])



**Step 13:** Create a simple report visual using fields from related tables.**Result**

Data was successfully extracted from the Global Superstore dataset, transformed using Power Query Editor, and loaded into the Power BI data model. Data cleaning, standardization, relationship creation, and DAX-based metric generation were performed successfully. The final dataset was prepared for reporting and business analysis.

**Learning Outcomes**

After completing this experiment, students will be able to:

- Explain the ETL process and its role in Business Intelligence systems. (CO1-BT2)
- Extract data from real-world business datasets using Power BI. (CO2-BT3)
- Apply transformation techniques including cleaning, filtering, and merging. (CO2-BT3)
- Create relationships and analytical measures in Power BI. (CO3-BT4)
- Validate transformed datasets for reporting and decision-making. (CO3-BT4)

**Text / Reference Books**

1. Sharda, Delen & Turban, Business Intelligence and Analytics: Systems for Decision Support.
2. Cindi Howson, Successful Business Intelligence.
3. Microsoft Power BI Official Documentation.
4. Microsoft Power Query Documentation.